Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

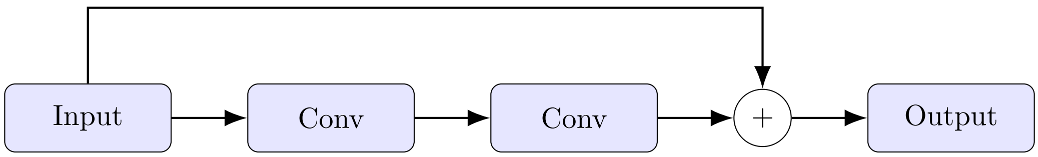

Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis

- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

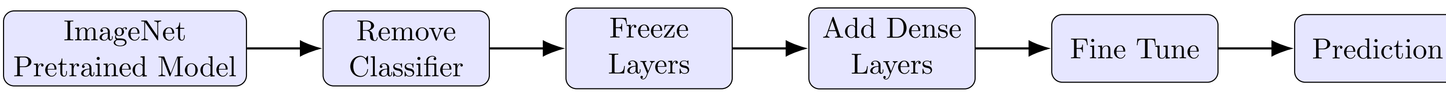

Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.

4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

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17. Mandatory Plots and Interpretations

The following plots were generated during the experiment. Each figure is followed by its corresponding interpretation.

17.1 Sample CIFAR-10 Images

Figure 3: Sample Images from the CIFAR-10 Dataset

Interpretation:

The sample images show examples from different CIFAR-10 classes. The images have a spatial resolution of 32×32 pixels and contain three color channels (RGB). The dataset contains visually diverse objects, which makes the classification task challenging for CNN models.

17.2 Training Accuracy and Validation Accuracy

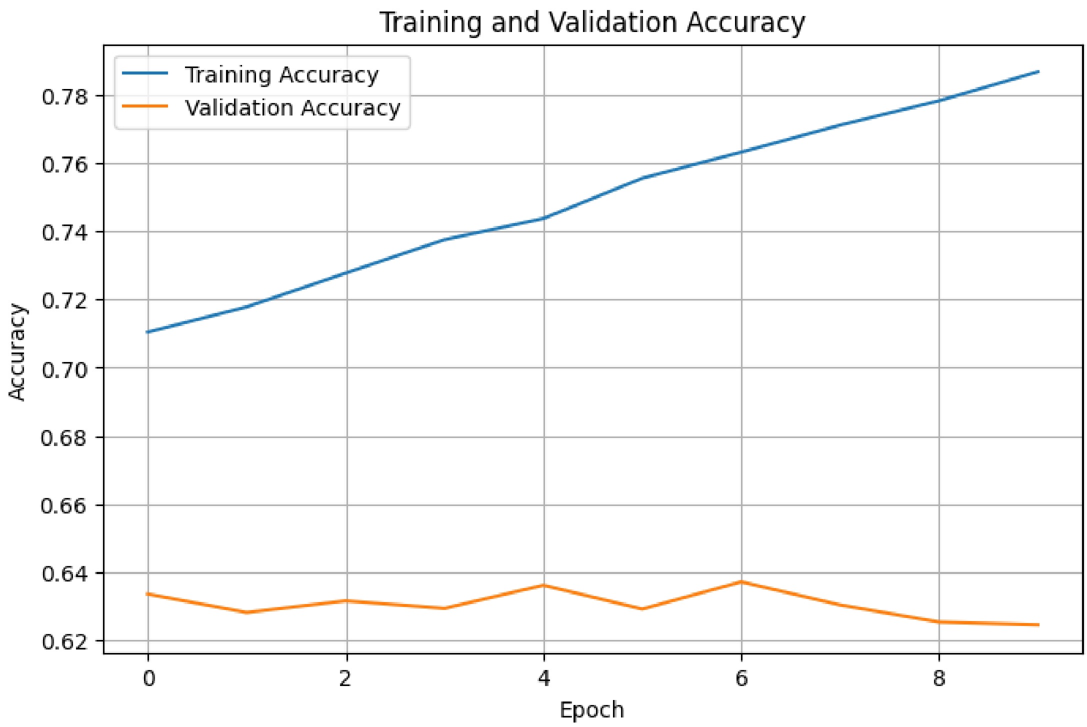

Figure 4: Training Accuracy of the Transfer Learning Model

Interpretation:

The training accuracy increases as the number of epochs increases, indicating that the model is learning useful features from the training dataset. of the model parameters. The validation accuracy is fluctuating and is finally decreasing after 6 epoch so the model may be overfitting.

17.4 Training Loss and Validation Loss

Figure 5: Training Loss of the Transfer Learning Model

Interpretation:

The training loss is decreasing as the epochs are increasing so the model predictions are becoming close to the actual data The Validation loss is increasing as number of epochs are increasing so the model is memorizing the data so the model may be overfitting.

17.6 Confusion Matrix

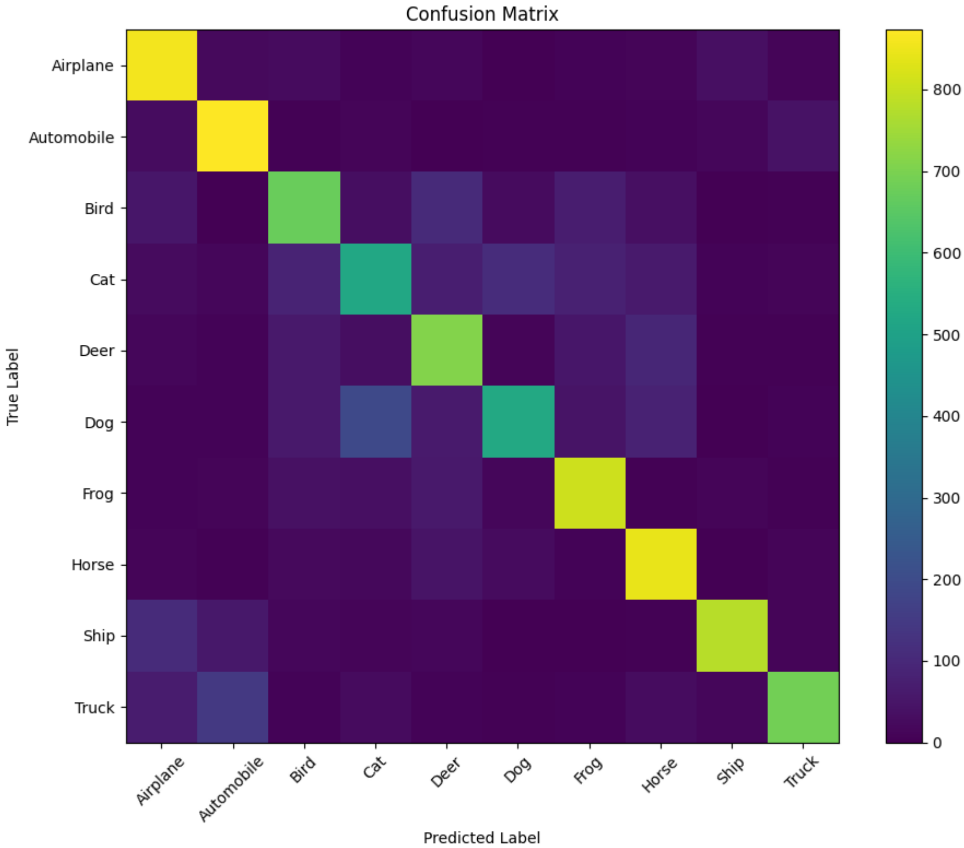

Figure 6: Confusion Matrix of the Classification Model

Interpretation:

The confusion matrix shows the number of correct and incorrect predictions for each CIFAR-10 class. There are higher values along the diagonal so the number of correct predictions are high.

18. Results

18.1 Performance Metrics

Metric	**Value**
Training Accuracy	78.69555354118347
Testing Accuracy	72.83%
Precision	73.16%
Recall	72.83%
F1-score	72.53%
Training Time	90ms
Total Parameters	14,848,586

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time (s)
LeNet-5	83,126	53.50	66.967267
AlexNet	36,925,322	10.00	341.690137
VGG16	14,848,586	10.00	700.675356
GoogleNet	995,082	80.01	270.104630
ResNet50	24,114,826	52.90	509.800042

19. Discussion Questions

Instructions: Answer each question based on the concepts studied and the observations obtained from the experiment.

Q1. Why is AlexNet considered a breakthrough in deep learning?

Answer: AlexNet was a breakthrough because it showed that deep CNNs could achieve very high accuracy on large image datasets. It used ReLU, GPUs, dropout, and data augmentation, which made deep learning much more effective for image recognition.

Q2. Why does VGG16 use only 3×3 convolution filters?

Answer:VGG16 uses 3×3 filters because they require fewer parameters and allow multiple convolution layers to be stacked, which helps the network learn more complex features.

Q3. Explain the advantages of the Inception module.

Answer:The Inception module uses different filter sizes in parallel. This allows the network to detect features at different scales while using computational resources efficiently.

Q4. What is the purpose of residual learning?

Answer:Residual learning allows a network to learn only the difference, or residual, between the input and desired output. Skip connections make it easier to train very deep networks and help reduce vanishing-gradient problems.

Q5. Differentiate LeNet and ResNet.

Answer:LeNet is a simple and shallow CNN designed mainly for handwritten digit recognition. ResNet is a much deeper CNN designed for complex image classification tasks. LeNet uses basic convolution, pooling, and fully connected layers. ResNet uses residual/skip connections to make deep networks easier to train. Therefore, LeNet is simpler and faster, while ResNet is more powerful but computationally expensive.

Q6. What is Transfer Learning?

Answer:Transfer learning is the process of using knowledge learned by a pretrained model on one task for another related task.

Q7. Why is fine tuning required?

Answer:Fine-tuning is required to adapt a pretrained model to the specific features of a new dataset.

Q8. Explain the difference between dilated convolution and transpose convolution.

Answer:Dilated convolution uses gaps between filter elements to increase the receptive field. It helps the network capture information from a larger area without greatly increasing parameters. Transpose convolution is mainly used to increase the spatial size of feature maps. For example, it can upsample a 16×16 feature map to 32×32. Therefore, dilated convolution expands the receptive field, while transpose convolution increases spatial resolution.

Q9. Why do pretrained models converge faster?

Answer:retrained models converge faster because they already contain useful features learned from a large dataset. Therefore, they do not need to learn basic features from random initialization.

Q10. Compare the computational complexity of LeNet and ResNet.

Answer:LeNet has fewer layers and parameters, so it requires less computation. ResNet is much deeper and contains many more convolution operations, so it requires significantly more computational resources.

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20. Additional Exercises

1. Implement transfer learning using VGG16.
2. Repeat the experiment using ResNet50.
3. Compare Adam and SGD optimizers.
4. Compare training with frozen layers and fine tuning.
5. Evaluate the model using Precision, Recall and F1-score.
6. Compare the performance of LeNet, AlexNet and ResNet.

21. References

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3. Karen Simonyan and Andrew Zisserman, *Very Deep Convolutional Networks for Large-Scale Image Recognition*, ICLR, 2015.
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5. Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
7. TensorFlow Documentation: `https://www.tensorflow.org`
8. Keras Documentation: `https://keras.io`

GitHub Repository:

`https://github.com/shaliniparvathareddy/Lab4_Deep_Learning`